

DES Pune University, Pune
School of Engineering and Technology
Department of Computer Engineering and Technology
Program: B. Tech in Computer Science and Engineering

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PRN No.: 1012411011

Name: Riddhee Kulkarni

Subject: Python Programming

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Lab Assignment: 23

Aim:

To write a program to open a file and print its attribute values.

Objective:

- To understand file handling in Python.
- To learn how to open and access files using built-in functions.
- To explore different file attributes such as name, mode, closed status, etc.
- To display file-related information effectively.

Theory:

File handling is an important concept in Python that allows users to work with files for reading, writing, and modifying data. Python provides built-in functions such as `open()` to handle files.

When a file is opened using the `open()` function, it returns a file object that contains various attributes.

These attributes provide useful information about the file. Some commonly used file attributes are:

- **name** – Returns the name of the file.
- **mode** – Returns the mode in which the file is opened (read, write, etc.).
- **closed** – Returns whether the file is closed or not (True/False).
- **readable()** – Checks if the file is readable.
- **writable()** – Checks if the file is writable.

By accessing these attributes, we can gather details about the file and its current state. Proper file handling also involves closing the file after operations using the `close()` method.

Algorithm:

1. Start the program.
2. Specify the file name.
3. Open the file using the `open()` function in appropriate mode.
4. Display the file name using the `name` attribute.
5. Display the file mode using the `mode` attribute.
6. Check whether the file is closed using the `closed` attribute.

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7. Check if the file is readable using readable().
8. Check if the file is writable using writable().
9. Close the file using the close() method.
10. Again check and display the closed status.
11. End the program.

Pseudo Code:

```
START
OPEN file in read mode
PRINT file name
PRINT file mode
PRINT whether file is closed
PRINT whether file is readable
PRINT whether file is writable
CLOSE file
PRINT whether file is closed
END
```

Code:

```
# Open a file
file = open("sample.txt", "r")

# Print file attributes
print("File Name:", file.name)
print("File Mode:", file.mode)
print("File Closed:", file.closed)

# Close the file
file.close()
# Check again after closing
print("File Closed after closing:", file.closed)
```

Output:

```
File Name: sample.txt
File Mode: r
File Closed: False
File Closed after closing: True

[Done] exited with code=0 in 0.595 seconds
```

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Conclusion:

Thus, the Python program was successfully executed to open a file and display its attribute values such as name, mode, readability, writability, and closed status. This helped in understanding file handling operations and how to access file properties efficiently in Python.